

Below is a good `plan.md` that you can place in the root of your Replit project. It is designed as an implementation roadmap that an AI coding agent (Replit Agent) can execute step-by-step.

It combines the existing HabitifyMe PRD, the build log, and the new Habit Formation Engine into a single V4 roadmap focused on **behavioral psychology, XP, retention, and habit formation**, instead of simply adding more features.

</> Markdown



```
# HabitifyMe V4 Implementation Plan
## Habit Formation Engine + XP System
```

Objective

Transform HabitifyMe from an AI workout planner into a true habit-building platform.

Current app:

- AI workout planning
- Adaptive weekly workouts
- Diet planning
- Workout logging
- Weekly review

New goal:

Build an addiction loop using

Trigger

→ Action

→ Reward

→ Investment

→ Repeat

based on BJ Fogg, Hook Model, Atomic Habits and the Habit Formation PRD.

North Star

Increase Week 4 completion rate.

Secondary metrics

- Day 2 retention
- Day 7 retention
- Day 14 retention
- Workout completion
- Weekly review completion
- Check-in frequency
- Habit Score improvement

- XP earned per week

IMPORTANT

Do NOT redesign the application.

Preserve

- Existing navigation
- Existing dark glass UI
- Existing workout generation
- Existing onboarding
- Existing workout pages

This project is an enhancement of the existing application.

Always reuse existing components before creating new ones.

Phase 1

Read Existing Project

First inspect

- PRD
- Current routes
- Components
- Database schema
- Existing Supabase functions

Understand

Current workout flow

Current weekly review

Current calendar

Current progress tracking

Current push notification implementation

Do not modify anything yet.

Phase 2

Build XP Engine

Create a centralized XP service.

Never calculate XP inside UI components.

Create

lib/xp.ts

Expose

awardXP()

deductXP()

getCurrentLevel()

getLevelProgress()

getXPHistory()

Every XP transaction must be saved.

Database

xp_transactions

Fields

id

user_id

reason

amount

created_at

metadata

XP Rules

Workout Complete

+50 XP

Gym Check-in

+20 XP

Level 5

1400 XP

Continue exponentially.

Every level unlocks

- celebration animation
- badge
- motivational copy

Phase 4

Habit Score

Create

Habit Score

0-100

Calculated nightly.

Weights

Workout Completion

40%

Gym Check-ins

25%

Diet Logging

20%

Weekly Review

15%

Rolling window

7 days

30 days

Store

habit_score

history

trend

Never compute inside frontend.

Phase 5

Habit Dashboard

Extend Home Screen

Show

Habit Score

Weekly XP

Current Level

XP Progress Bar

Today's Habit Status

Consistency

Weekly graph

Recent achievements

Keep visual hierarchy minimal.

Phase 6

Gym Check-In

Add

Quick Check-In button

Home screen

One tap.

Records

time

location (optional)

date

Feeds

Habit Score

XP

Retention metrics

Do NOT require workout logging.

Phase 7

Trigger Engine

Enhance onboarding.

Capture

Preferred workout time

Workout days

Reminder preference

Timezone

Notification strategy

30 min before workout

Missed workout reminder

3 day recovery

7 day restart

14 day restart

Fresh Week restart

Phase 8

Recovery Flow

If inactive

3 days

Offer

15 minute workout

Easy XP

Motivation

Ask

"What prevented today's workout?"

Capture

Free text

Reason category

Store analytics.

Phase 9

Sunday Planning

Every Sunday evening

Prompt

Ready for next week's plan?

Allow

Travel mode

Unavailable days

Regenerate plan

Award XP

Phase 10

Surprise Rewards

Implement randomness

10% chance

Reward

Bonus XP

Rare badge

Confetti

Never predictable.

Phase 11

Achievements

Create badge engine

Examples

First Workout

3 Day Streak

7 Day Streak

Week Complete

10 Workouts

1000 XP

Perfect Week

Habit Champion

Badges are permanent.

Phase 12

Offline Support

Cache

Current workout

Current diet

Weekly plan

Allow logging offline.

Sync later.

Phase 13

Analytics

Track

notification_opened

checkin_completed

workout_started

workout_completed

xp_awarded

badge_unlocked

habit_score_updated

weekly_review_submitted

plan_regenerated

restart_clicked

recovery_prompt_opened

Store events.

Phase 14

Segmentation

Automatically classify users

0-20

At Risk

Unlock

Easy workouts

XP multipliers

Recovery prompts

21-60

Building

Normal experience

61-80

Strong Habit

Advanced plans

Leaderboards

81-100

Champion

Referral

Fit Squad

Exclusive badges

Phase 15

UI Animations

Small delightful moments

XP popup

+50 XP

Confetti

Level up

Badge unlock

Habit Score increase

Never overuse animations.

Phase 16

Database

Create

xp_transactions

habit_scores

habit_score_history

checkins

achievements

notification_preferences

user_settings

analytics_events

Indexes

user_id

created_at

Phase 17

Edge Functions

Create Supabase Edge Functions

awardXP

calculateHabitScore

sendReminder

weeklyPlanner

recoveryPrompt

achievementEngine

Keep business logic server-side.

Phase 18

Refactoring

Remove duplicated XP calculations.

Extract

Reward Engine

Habit Engine

Notification Engine

Analytics Engine

Achievement Engine

Phase 19

Testing

Test

Workout completion

XP

Badges

Habit Score

Recovery flow

Notifications

Offline mode

Edge cases

No duplicate XP

No negative score

No multiple rewards

Phase 20

Final QA

Verify

Existing PRD still works.

Existing Build Log fixes remain intact.

No regression.

Performance remains smooth.

Mobile responsiveness maintained.

Coding Rules

Always:

- ✓ Reuse existing components
- ✓ Keep dark glass design
- ✓ Keep TypeScript strict
- ✓ Use Supabase best practices
- ✓ Prefer composable services
- ✓ Server-side calculations
- ✓ Small reusable hooks

Avoid

- ✗ Massive components
- ✗ XP logic inside UI
- ✗ Duplicate database queries
- ✗ Breaking existing navigation
- ✗ Hardcoded values
- ✗ Business logic inside React components

Deliverables

1. XP Engine
2. Habit Score Engine
3. Achievement System
4. Check-in System
5. Trigger Engine

6. Recovery Flow

7. Sunday Planning

8. Analytics

9. Offline Support

10. Dashboard Enhancements

11. Edge Functions

12. Database Migrations

13. Tests

14. Documentation

The goal is to create a sticky, habit-forming experience that improves user retention thr

This plan is aligned with your Habit Formation document's trigger-action-reward-repeat model, extends the existing HabitifyMe roadmap rather than replacing it, and respects the architecture and completed work documented in the PRD and build log.